



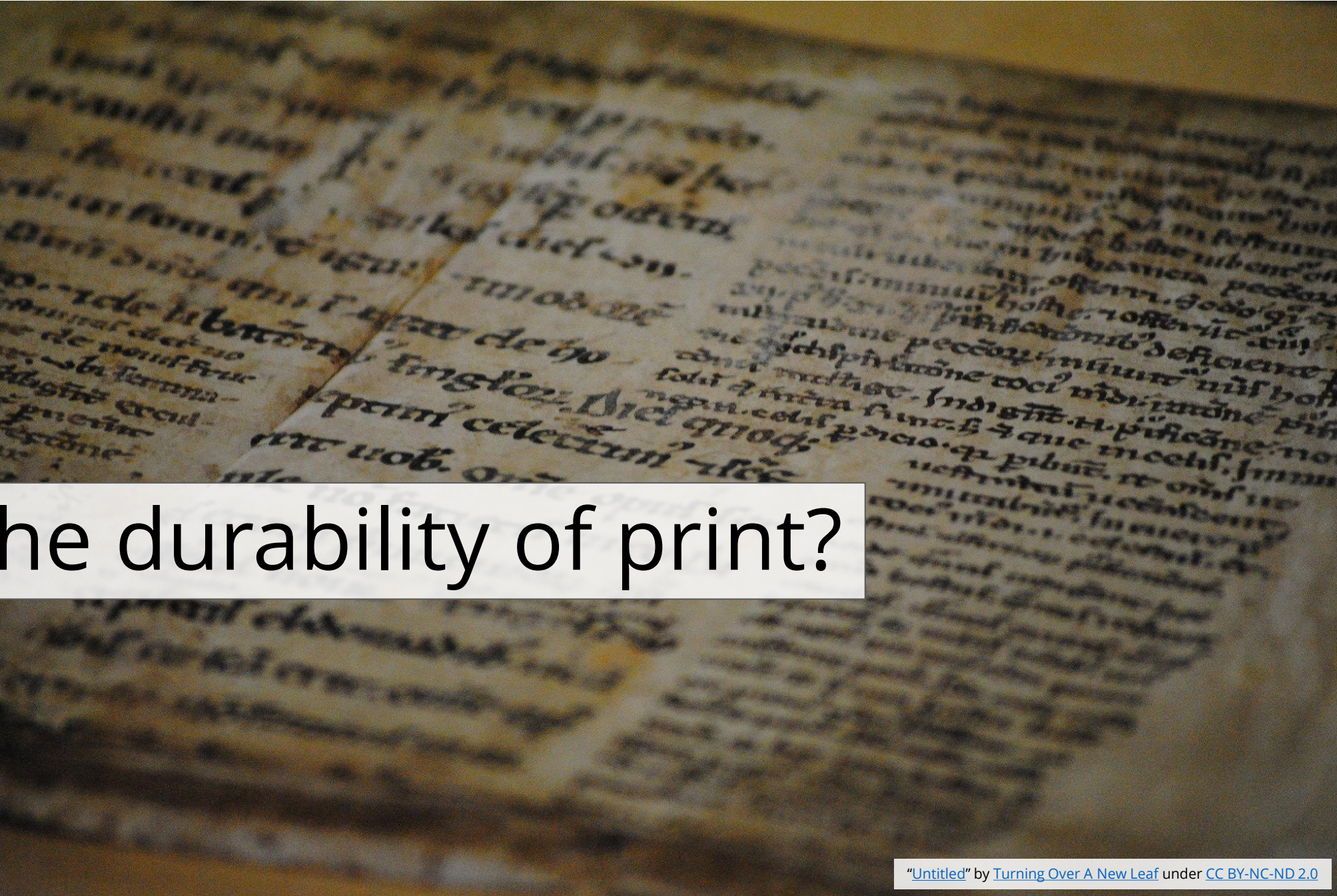
Introduction to Personal Digital Archiving

Nicholas Taylor
nullhandle.org

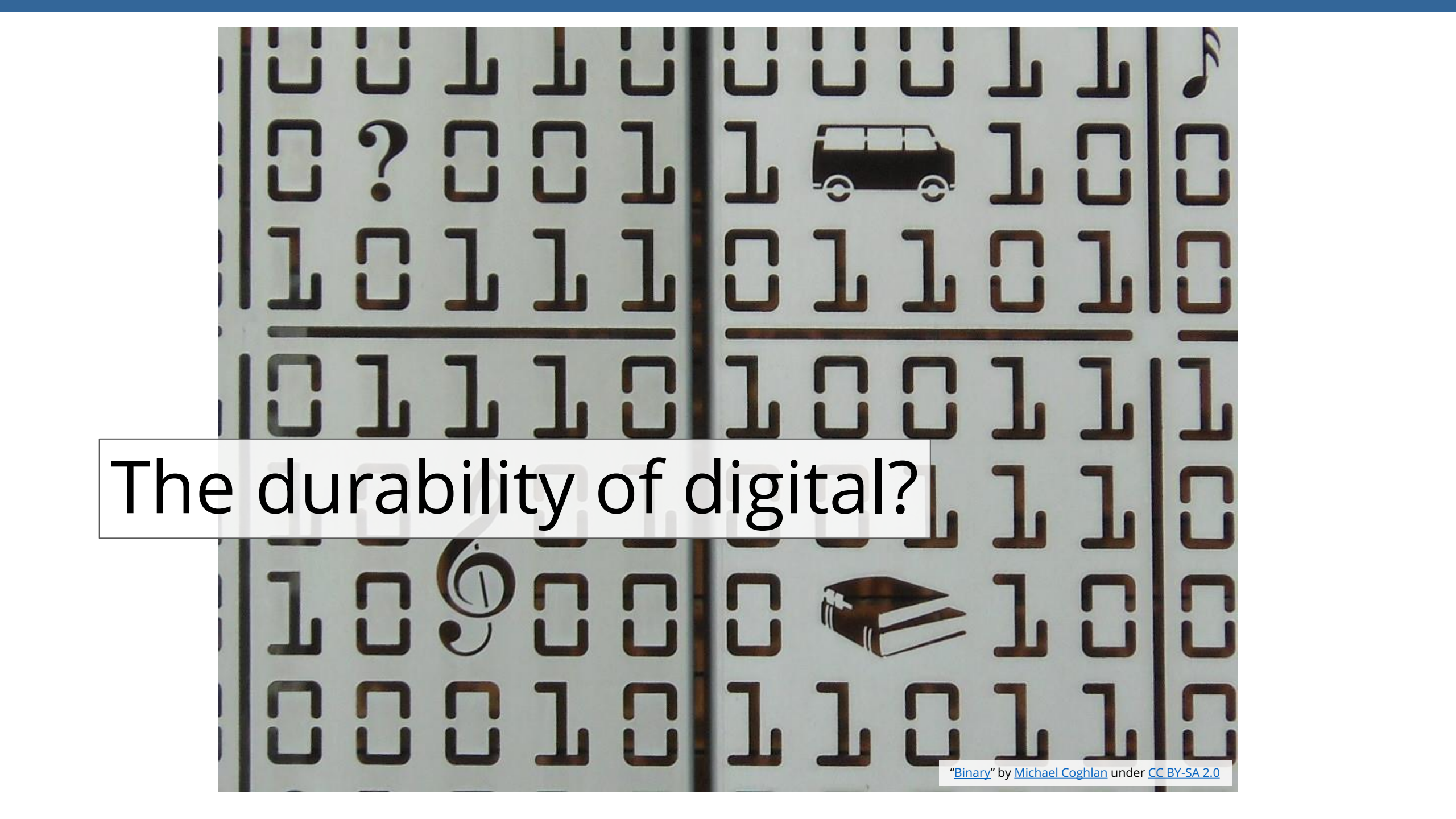
[Santa Fe Community College Continuing Education](#)
27 September + 4 October 2026



My Grandmother's Archive



The durability of print?



The durability of digital?

What is *personal digital archiving*?

“The practice of preserving one’s own digital records of continuing value.”

– [Dictionary of Archives Terminology, Society of American Archivists](#)



What is *personal digital archiving*?

In other words, *personal digital archiving* is a set of practices aimed at **enhancing the utility and longevity of personal digital artifacts**.



"VPHS Boxes" by [Taber Andrew Bain](#) under [CC BY 2.0](#)

Goals for this Course

- Equip you to be **more confident and effective stewards** of your personal digital artifacts
- **Think critically and intentionally** about what you'll save and curate, why, and for whom





What other goals do you have for this course?

Overview

- Background
- Inventory
- Appraise
- Organize
- Preserve
- Access



Background



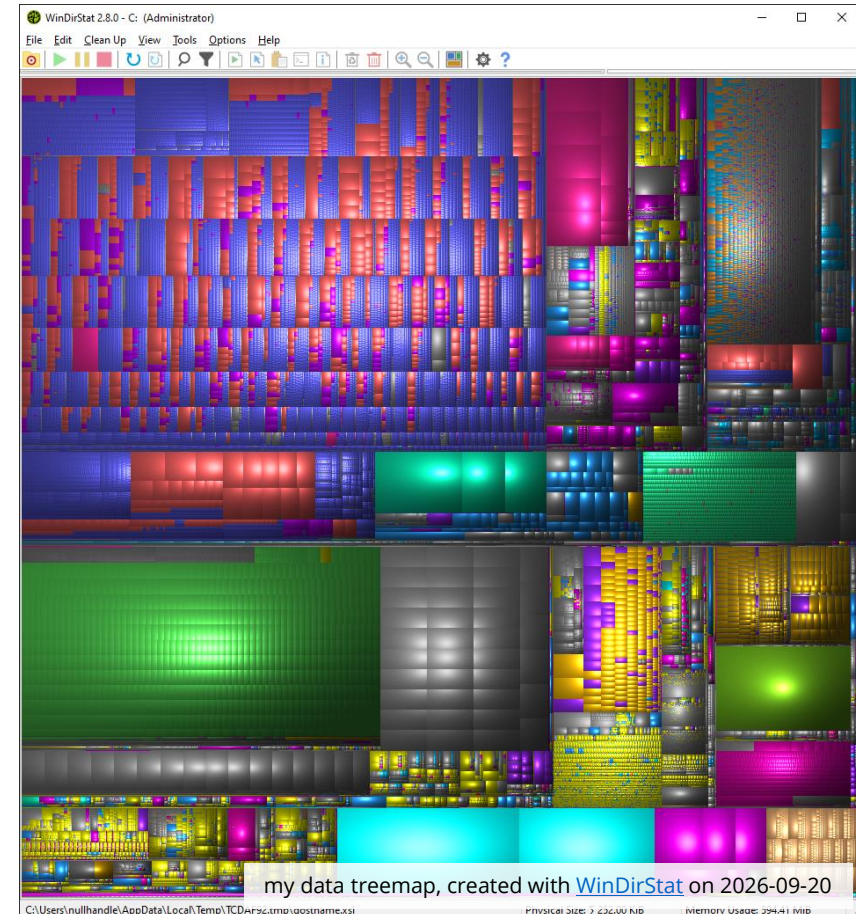
My expertise + background

- Librarian + expert witness
- Work on digital libraries, digital preservation, web archiving
- 18+ years in research libraries + tech firms
- Google, Sun, SCOTUS, Library of Congress, Stanford, LANL
- M.L.S., UMD / M.A., Comm., Culture, & Tech., Georgetown
- Website: nullhandle.org

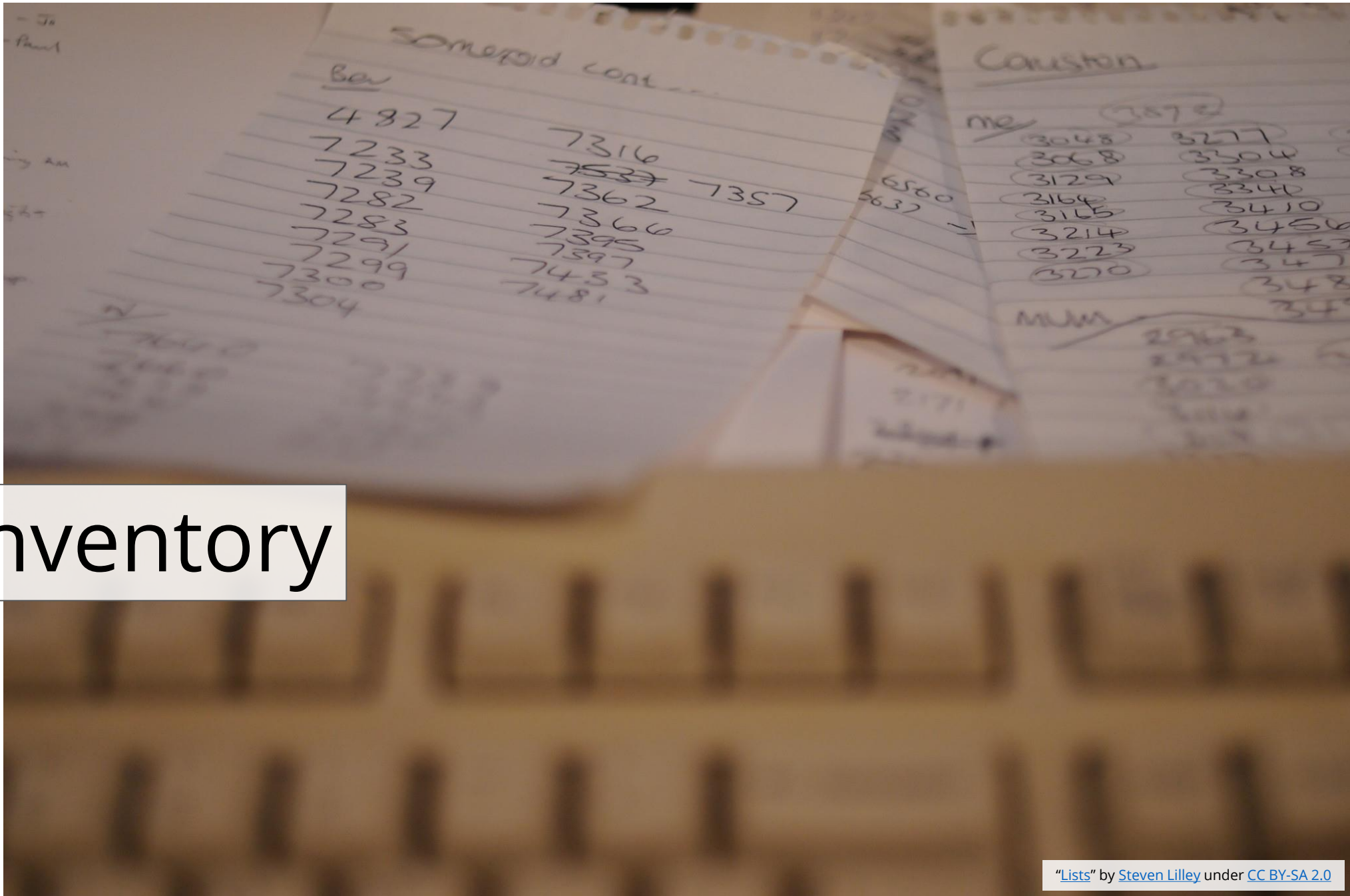


My personal digital archive

- Born-digital photos, 1999-
- Born-digital video, 2000-
- Digitized photos, 1979-1998
- E-mails, 1996-
- School work, 1997-
- Presentations, 2009-
- Genealogy files, 2001-
- Websites, 2001-



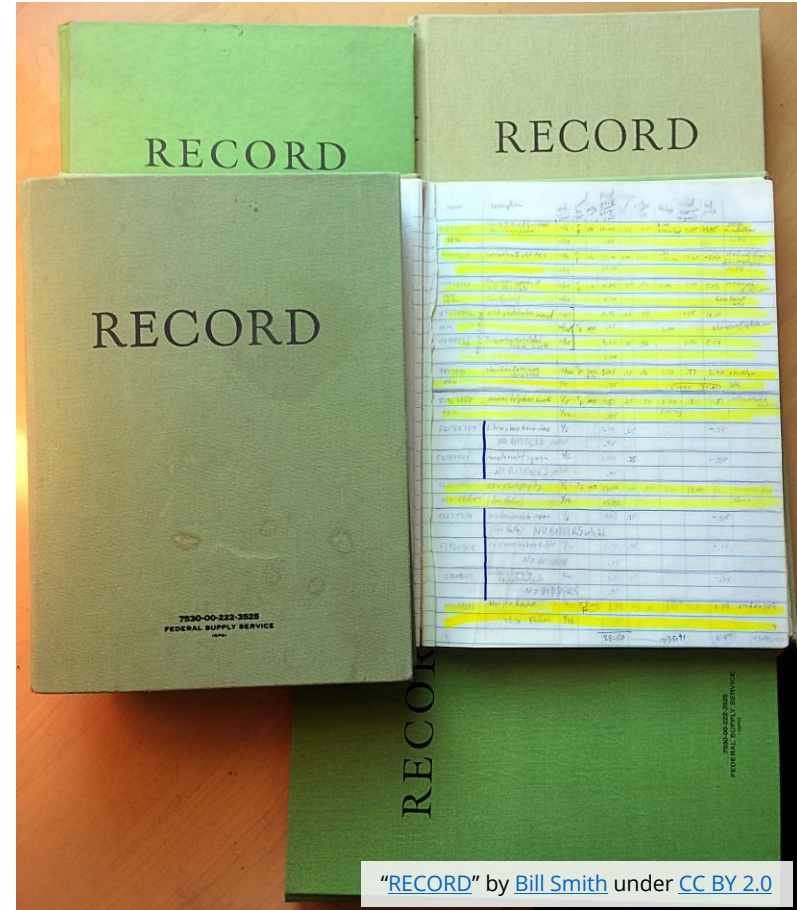
Inventory



What do we mean by *inventory*?

“The process of documenting the records in a location, typically at the series level.”

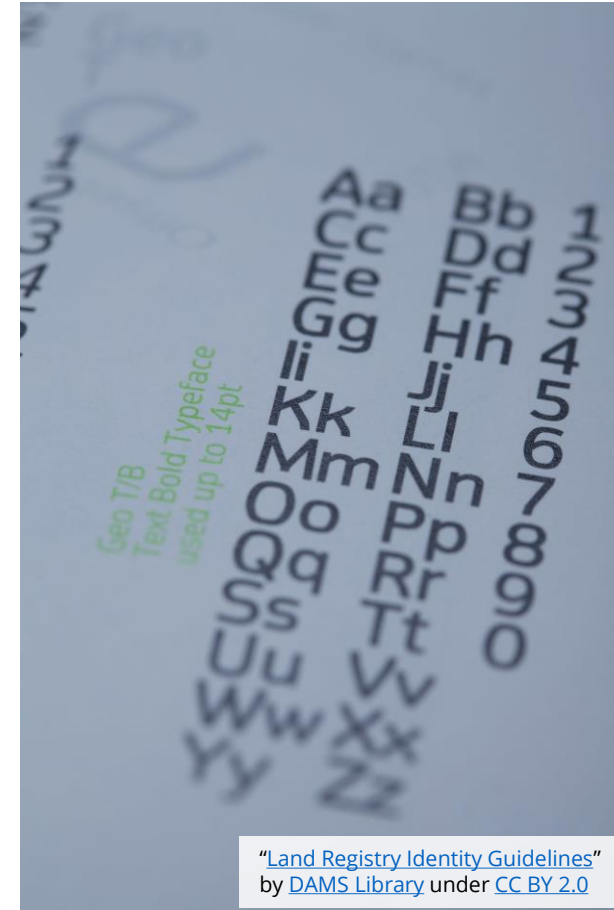
– [Dictionary of Archives Terminology, Society of American Archivists](#)



What do we mean by *inventory*?

In other words, to *inventory* is to systematically canvass, then document, **what assets are stored where**.

When inventorying, capture the minimum level of detail that will **support subsequent appraisal and organization**.





What are some places where personal digital artifacts of interest may be found?

Places Our Data Lives (and, Well, Dies)

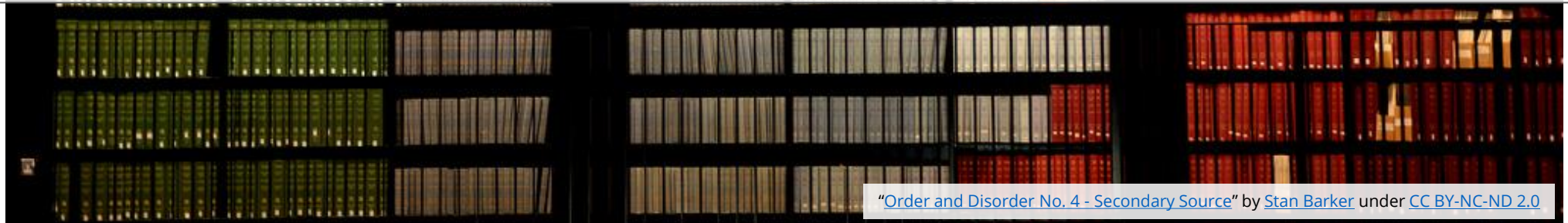
- Legacy media (floppy disks, Zip disks, CDs, DVDs)
- Flash drives
- Hard drives (internal, external)
- Cloud storage
- (Hosted) e-mail
- Smartphone
- Camera



"Selection of obsolete media at the 2010 National Book Festival" by [wlef70](#) under [CC BY-NC-SA 2.0](#)



What are your most and least controlled digital artifacts?



Practical Step: Inventory Storage

- Desktops + laptops
- Phones + tablets
- Removable media
- Cloud services



"Hard Drives" by Gregg Tavares under CC BY 2.0

Practical Step: Inventory Data

Basic Characteristics

- What kinds of data, where?
- How much data, where?

Risk Assessment

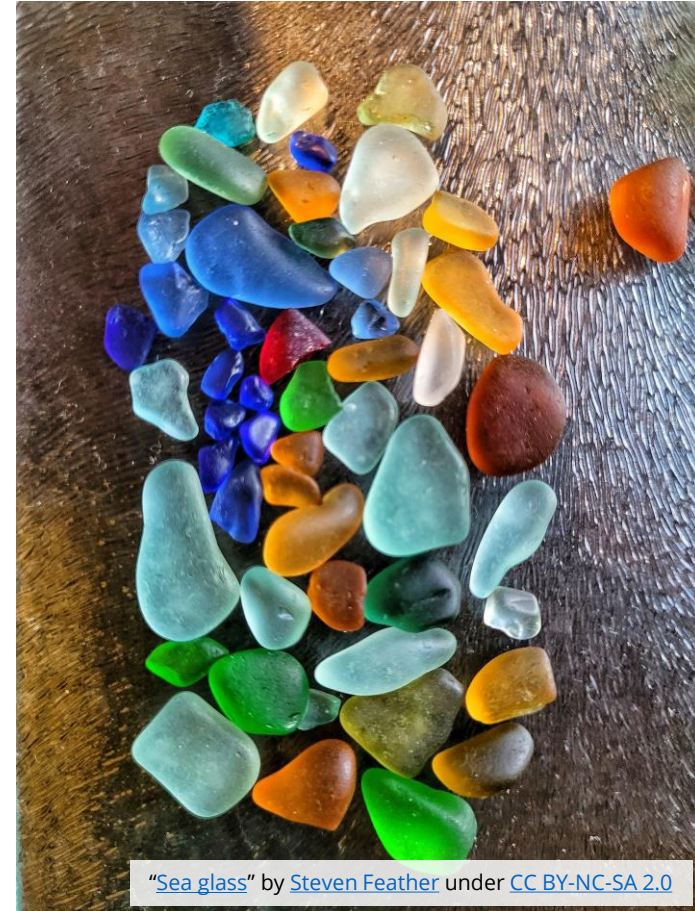
- What data is least redundant?
- What data is most vulnerable?
- What data depends on particular software?

Data Flows

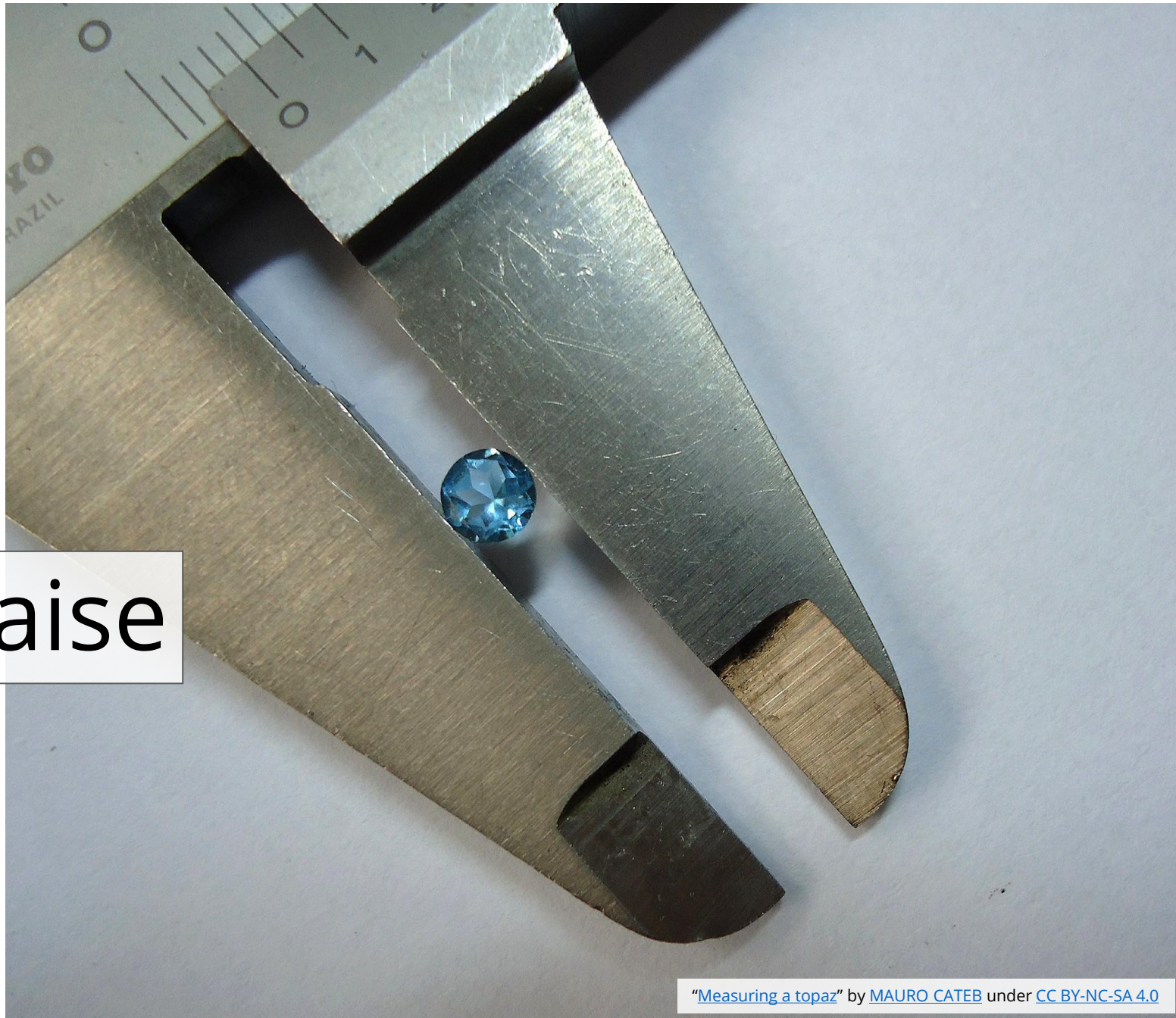
- Where is new data created?
- Is data duplicated?
- Where does canonical data live?

Common Data of Interest

- Photos
- E-mail
- Documents
- Genealogical data
- Social media + web(?)



Appraise



What do we mean by *appraise*?

“The process of determining whether records and other materials have permanent (archival) value.”

- [Dictionary of Archives Terminology, Society of American Archivists](#)

“The basis of appraisal decisions may include a number of factors, including the records’ provenance and content, their authenticity and reliability, their order and completeness, their condition and costs to preserve them, and their intrinsic value.”

What do we mean by *appraise*?

In other words, to *appraise* is to make a determination about **the threshold value of an artifact**.

Appraisal informs **decision-making about resource investments**.



"frAMed" by [Robert Hruzek](#) under [CC BY-NC-ND 2.0](#)



What are some criteria by which you ascribe value to personal digital artifacts?

What resource investments?

The backdrop of personal digital archiving (and, well, everything else) is **finite resources**.

Time and money that we expend on our archives **trade off against other investments**.



“Good enough” is great

We ideally want to align the level of effort expended curating with the **expected needs for access**.

Curate to the level that supports how and what you and others **want to be able to find and explore**.



Professional Archivists' "Secret"

Not everything is worth saving.

Archival orgs' sustainability depends heavily on **their selectivity**.

In the scheme of all that could be preserved, archival orgs can only save **a vanishingly small sliver**.

It's admittedly more challenging in the digital context, where the **externalities of accumulation** are less conspicuous.



"Discards" by [haunted by Leonard Cohen](#) under [CC BY-SA 2.0](#)



What are your most prized digital artifacts?

Practical Step: Appraisal

- Review your inventory
- Consider resource constraints
- Decide what's worth preserving
- Decide what's worth curating and to what level
- Decide what to discard(?)



Practical Step: Appraisal

Resources

- How do resource constraints (e.g., in time, money, space) inform determinations of the relative value of digital artifacts?

Heuristics

- What heuristics can you develop to determine if you will keep a given digital artifact?
- What heuristics can you develop to determine if you will curate a given digital artifact, to what level?

Who cares?

- Who else cares about artifacts in your personal digital archive?
- Who else could care about the artifacts in your personal digital archive (i.e., for whom should the possibility of the archive's future relevance be stewarded)?

Who cares? Who could care?

Both your ancestors and your descendants are **stakeholders in the legacy you steward.**

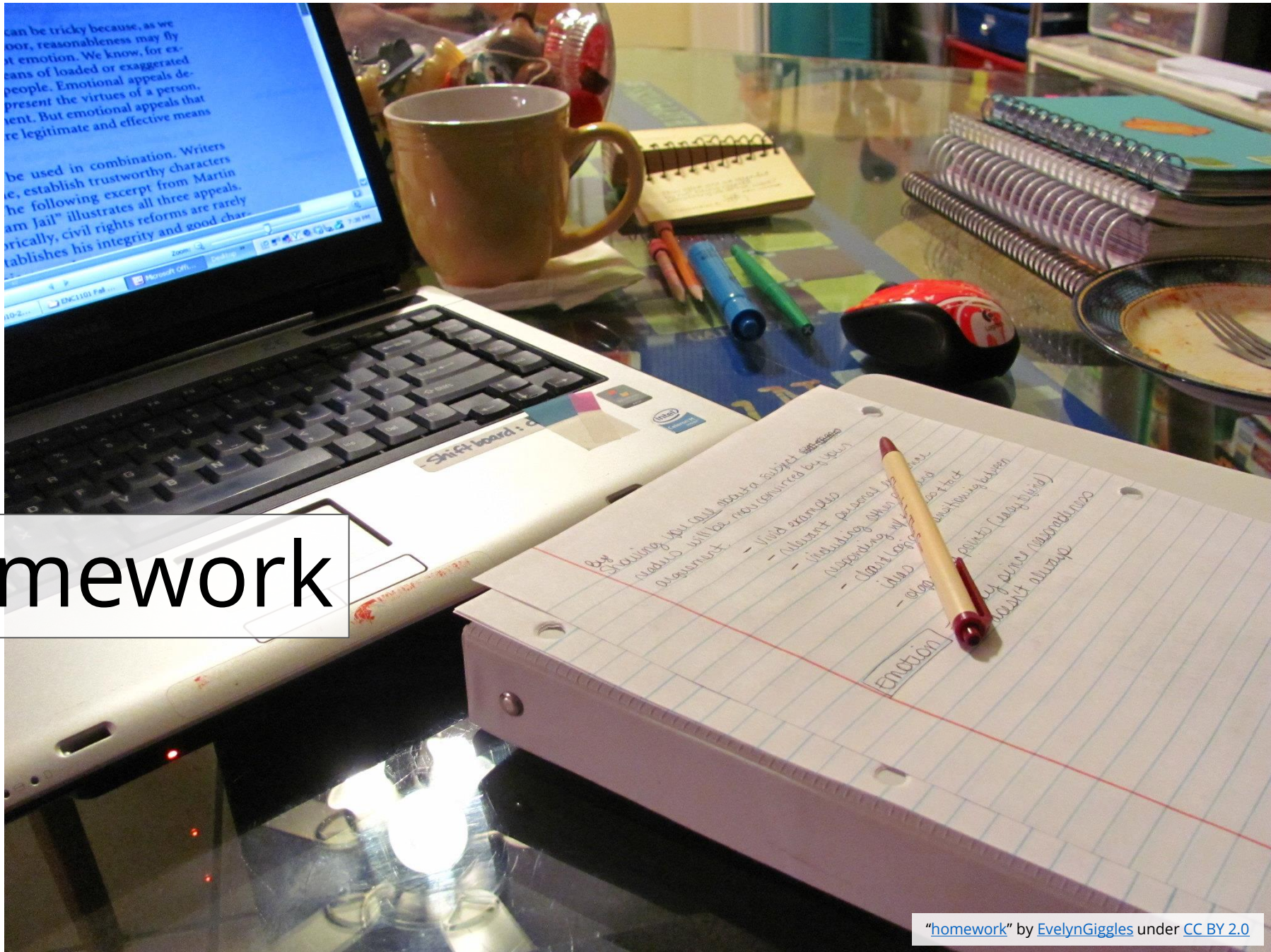
Ancestors will have more preconceptions of what that means, but I recommend that you **weight your actions based on your descendants.**

Mind that a well-curated archive is **both a gift and a debt.**



"Cothran Family Quilt" by [bonstance](#) under [CC BY-NC-SA 2.0](#)

Homework





Organize

What do we mean by *organize*?

Arrangement

“Organizing materials with respect to their provenance and original order, to protect their context and to achieve physical or intellectual control over the materials.”

– [Dictionary of Archives Terminology, Society of American Archivists](#)

Description

“Creating a set of data representing an archival resource or component thereof.”

– [Dictionary of Archives Terminology, Society of American Archivists](#)

What do we mean by *organize*?

Arrangement

In other words, *arrangement* is organizing digital artifacts in such a way as **to conserve their context and provenance.**

Description

In other words, *description* is creating and attaching metadata to digital artifacts **to improve their discoverability and interpretability.**



How do you organize your personal digital artifacts?

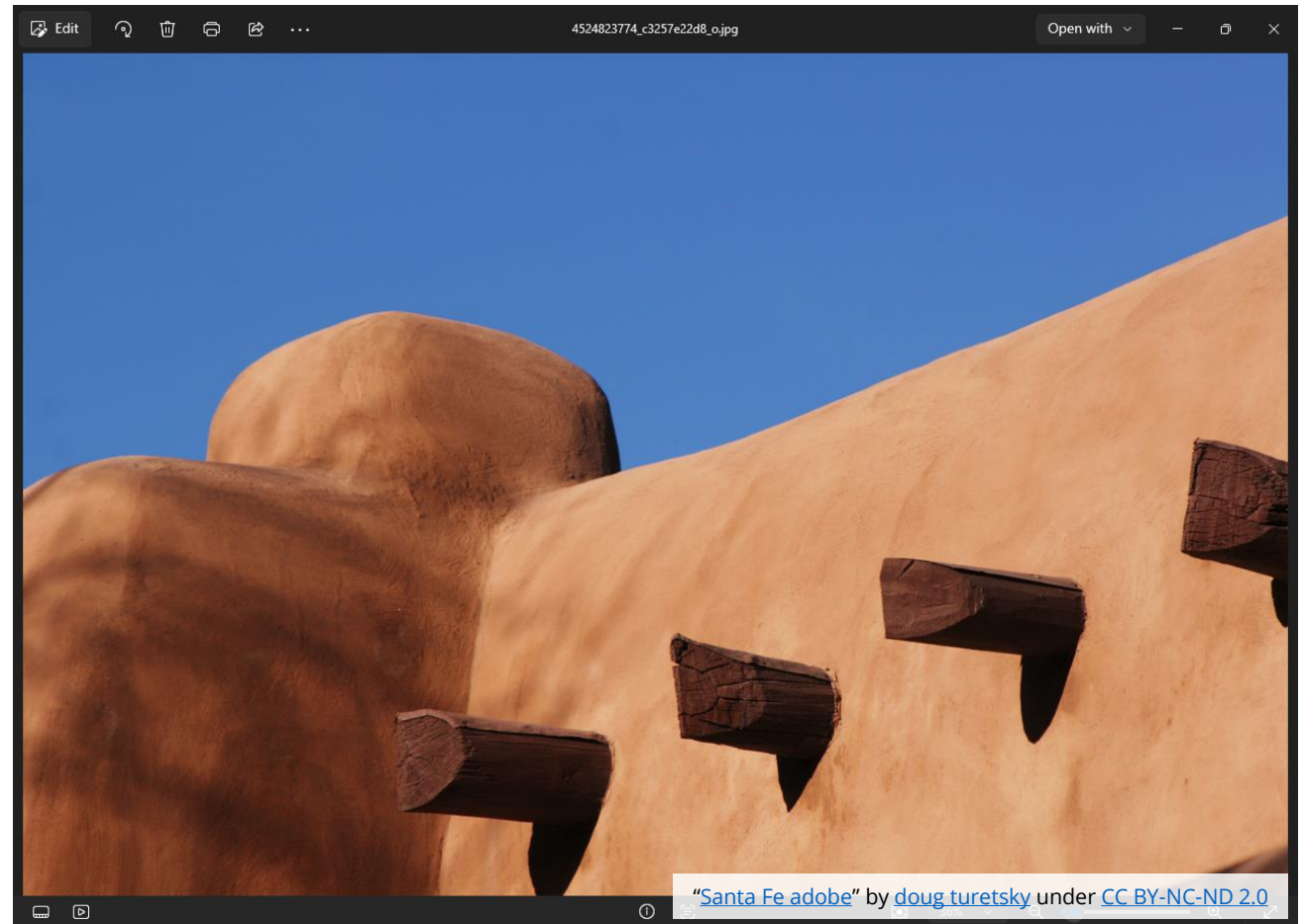


What purpose does that organizing support?

A Digital Photo in Microsoft Photos

Microsoft Photos is the basic bundled photo viewer application in Windows.

It facilitates quick viewing of photos and simple editing.



Same Photo in Notepad++

Notepad++ is a text editor.

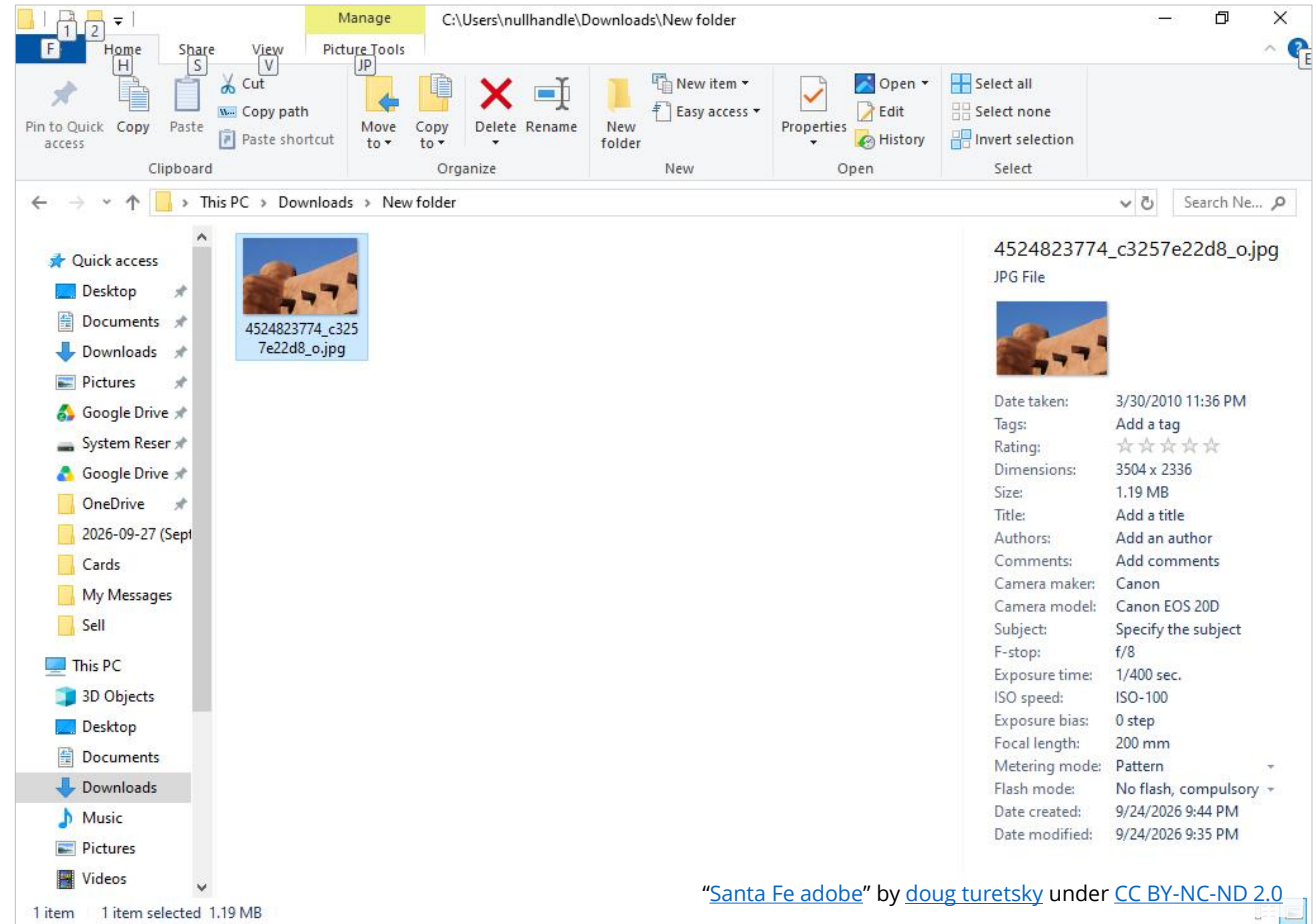
Obviously, it's not designed to display photos, but it can make sense of text embedded within the digital photo file – in this case, EXIF metadata including timestamps, the camera make and model, and software subsequently used to edit the photo.

```
1 y0yaNUTD3JFIFNUTSOHSTXSOH, SOH, NUTNUTy4DC2SOHExifNUTNUTII* NUTBSNUTNUTNUT
2 NUTSTSOHSTXNUTACRNUTNUTNUT: NUTNUTNUTD3SOHSTXNUTSONUTNUTNUT
3 NUTSTSOHSTXNUTACRNUTNUTNUT: NUTNUTNUTD3SOHSTXNUTSONUTNUTNUT
4 NUTSTSOHSTXNUTACRNUTNUTNUT: NUTNUTNUTD3SOHSTXNUTSONUTNUTNUT
5 NUTSTSOHSTXNUTACRNUTNUTNUT: NUTNUTNUTD3SOHSTXNUTSONUTNUTNUT
6 NUTSTSOHSTXNUTACRNUTNUTNUT: NUTNUTNUTD3SOHSTXNUTSONUTNUTNUT
7 NUTSTSOHSTXNUTACRNUTNUTNUT: NUTNUTNUTD3SOHSTXNUTSONUTNUTNUT
8 NUTSTSOHSTXNUTACRNUTNUTNUT: NUTNUTNUTD3SOHSTXNUTSONUTNUTNUT
9 NUTSTSOHSTXNUTACRNUTNUTNUT: NUTNUTNUTD3SOHSTXNUTSONUTNUTNUT
10 NUTSTSOHSTXNUTACRNUTNUTNUT: NUTNUTNUTD3SOHSTXNUTSONUTNUTNUT
11 NUTSTSOHSTXNUTACRNUTNUTNUT: NUTNUTNUTD3SOHSTXNUTSONUTNUTNUT
12 NUTSTSOHSTXNUTACRNUTNUTNUT: NUTNUTNUTD3SOHSTXNUTSONUTNUTNUT
13 NUTSTSOHSTXNUTACRNUTNUTNUT: NUTNUTNUTD3SOHSTXNUTSONUTNUTNUT
14 NUTSTSOHSTXNUTACRNUTNUTNUT: NUTNUTNUTD3SOHSTXNUTSONUTNUTNUT
15 NUTSTSOHSTXNUTACRNUTNUTNUT: NUTNUTNUTD3SOHSTXNUTSONUTNUTNUT
```

Same Photo in Windows Explorer

Explorer is the filesystem browser on the Windows operating system.

It can show us more of the EXIF metadata we observed when we opened the file in a text editor as well as some file metadata, but here the values are fielded so we can make better sense of them.

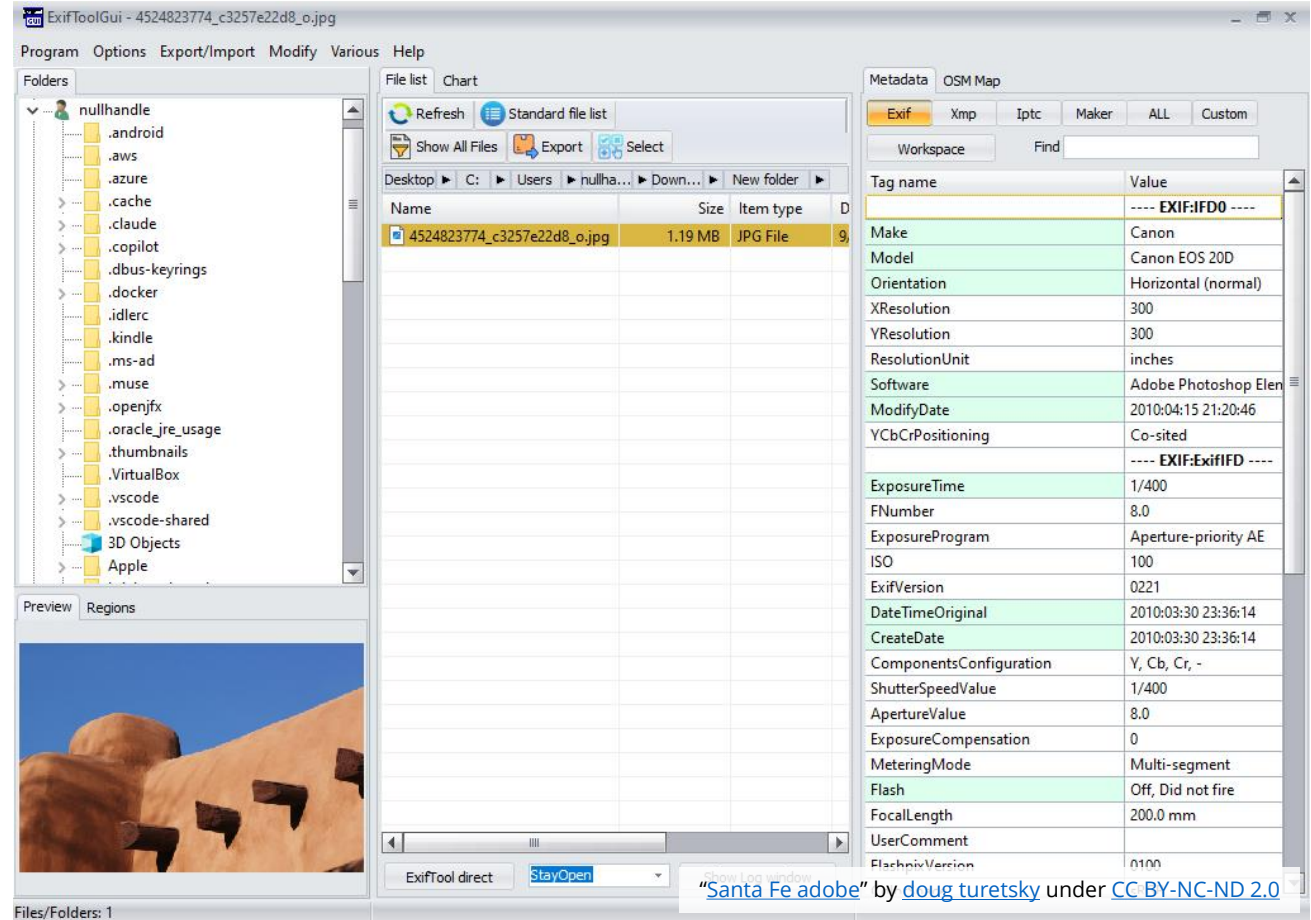


"Santa Fe adobe" by [doug turetsky](#) under [CC BY-NC-ND 2.0](#)

Same Photo in ExifToolGUI

ExifToolGUI facilitates viewing and editing embedded digital photo metadata.

It presents an even larger array of fields and field values.



The screenshot displays the ExifToolGUI application window. The left pane shows a file explorer with a list of folders. The center pane shows a file list with the selected file '4524823774_c3257e22d8_o.jpg' (1.19 MB, JPG File). The right pane shows the metadata for the selected file, organized into sections: Exif, Xmp, Iptc, Maker, ALL, and Custom. The 'Exif' section is active, showing a table of metadata fields and their values.

Tag name	Value
----- EXIF:IFD0 -----	
Make	Canon
Model	Canon EOS 20D
Orientation	Horizontal (normal)
XResolution	300
YResolution	300
ResolutionUnit	inches
Software	Adobe Photoshop Elements 2.0
ModifyDate	2010:04:15 21:20:46
YCbCrPositioning	Co-sited
----- EXIF:ExifIFD -----	
ExposureTime	1/400
FNumber	8.0
ExposureProgram	Aperture-priority AE
ISO	100
ExifVersion	0221
DateTimeOriginal	2010:03:30 23:36:14
CreateDate	2010:03:30 23:36:14
ComponentsConfiguration	Y, Cb, Cr, -
ShutterSpeedValue	1/400
ApertureValue	8.0
ExposureCompensation	0
MeteringMode	Multi-segment
Flash	Off, Did not fire
FocalLength	200.0 mm
UserComment	
FlashpixVersion	0100

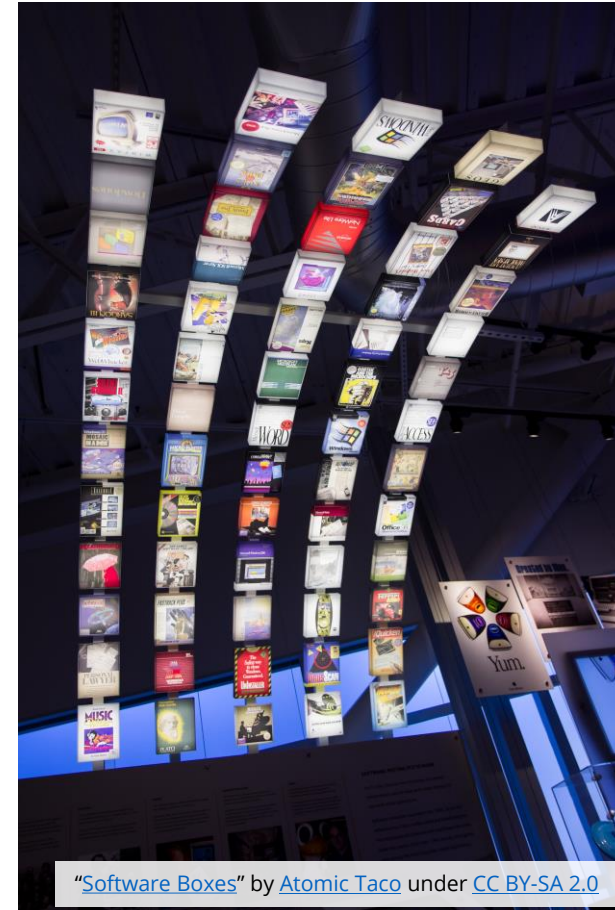
At the bottom of the window, there is a 'Show Log window' button and a text box containing the text: "Santa Fe adobe" by doug turetsky under CC BY-NC-ND 2.0.

The Role of Software

Software is **all but essential** to managing personal digital archives.

Different applications let you work with digital files **in different ways**.

Mind those capabilities and choose software that **serves your curation goals**.



Cloud Software Trade-Offs

Upsides

- Simplifies access
- Provides for social engagement w/ artifacts
- More cutting-edge curatorial tools
- Comes w/ native data redundancy

Downsides

- Your data is governed by unilaterally modifiable terms of service
- Your data may not be fully portable
- Your data may be used to train AI
- Data persistence contingent on factors beyond your control

Artificial Intelligence (AI)

AI is increasingly capable and can be used to great effect **for both organizing and exploring** personal digital archives.

Use **locally installed software** with integrated AI to keep your data from being used for commercial AI training.

Mind the **evolving capabilities** of different AI tools.



Data Structures: File vs. Database

File

An **atomic digital artifact** – e.g., a photo or document.

Different software allows for the **interaction with different files in different ways** – e.g., read and / or write.

Database

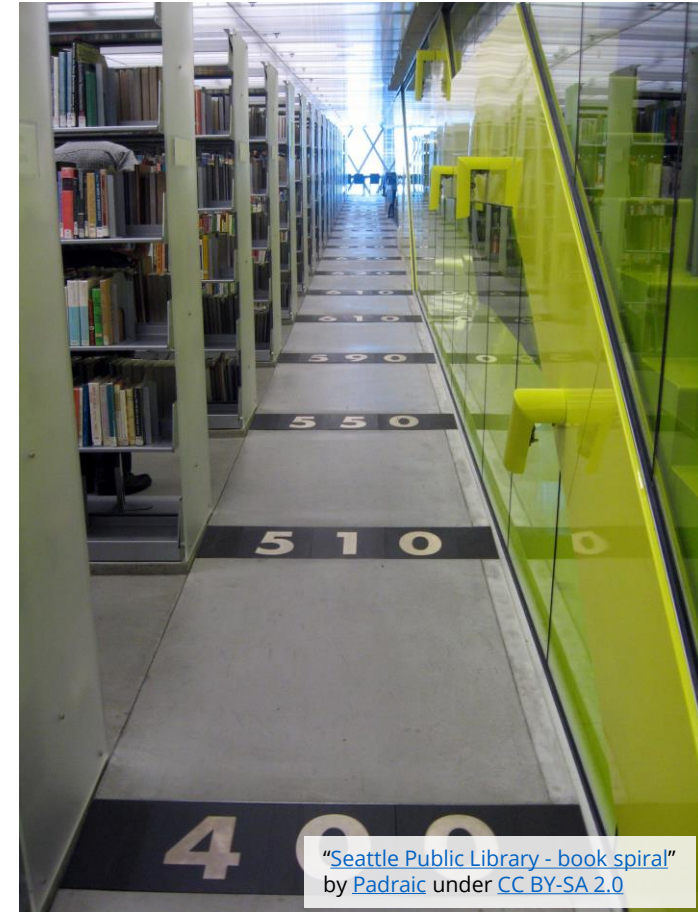
A database may store many files or information that might otherwise **be represented differently in many files**.

It supports **more flexible but also more complex** means of retrieving the data it stores.

What is *metadata*?

“Information about data that promotes discovery, structures data objects, and supports the administration and preservation of records.”

– [Dictionary of Archives Terminology, Society of American Archivists](#)



What is *metadata*?

In other words, metadata is **information about data**.

Annotations, captions, labels, keywords, description of any sort **are all examples of metadata**.

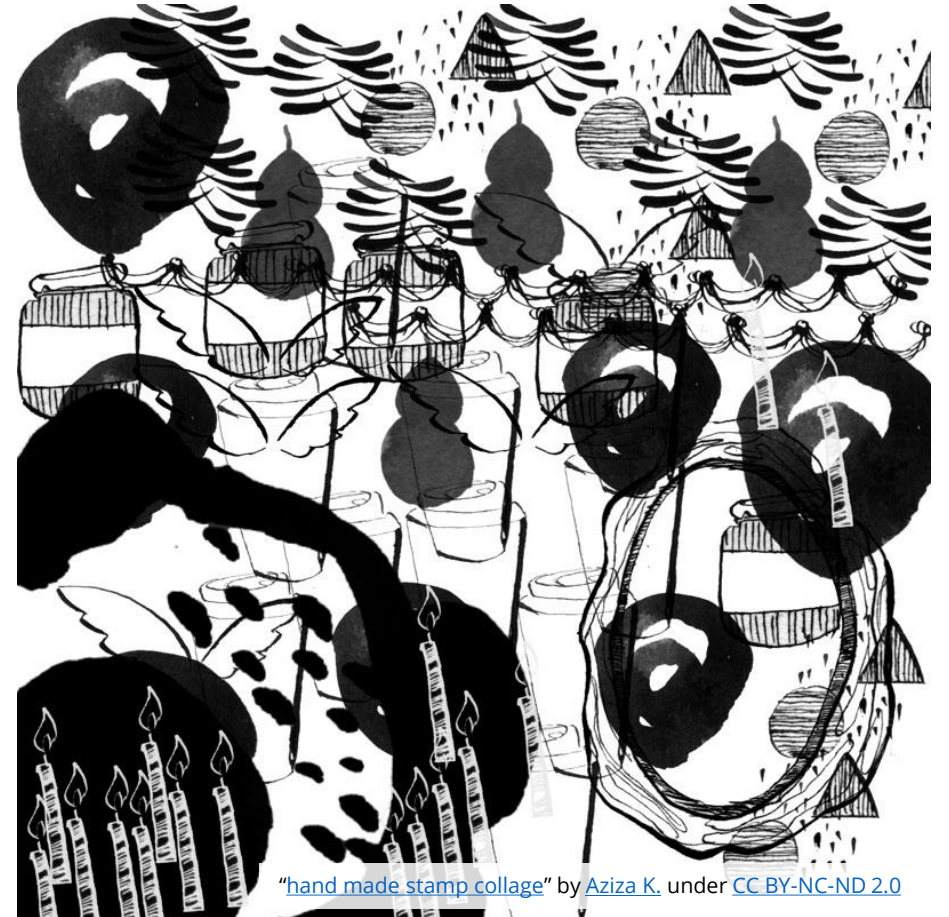
Metadata helps us **retrieve and understand** our personal digital artifacts.



Metadata Sources

Metadata can be an **incidental byproduct** of hardware and software's interaction with digital files.

It may also be **intentionally created** by people, to support the discovery and interpretability of data.



"hand made stamp collage" by [Aziza K.](#) under [CC BY-NC-ND 2.0](#)

How much metadata is enough?

Any is good; more is better.

What level of description is sufficient to **provide the opportunity** for someone to understand, connect with, and care about what the digital artifacts represent?



Metadata Standards

Metadata is of **sufficient commercial and societal importance** that stakeholders have invested in the development of metadata standards.

In a personal context, using metadata standards helps to ensure **portability and interoperability**.



"Weights and measures" by
Peter Burka under CC BY-SA 2.0



What kinds of data are you most invested in organizing?



How do you involve others in curating the archive?



Files and Folders

The *inventory* and *appraisal* steps inform folder structure.

First consider the **grossest levels of categorization**, then work your way down.

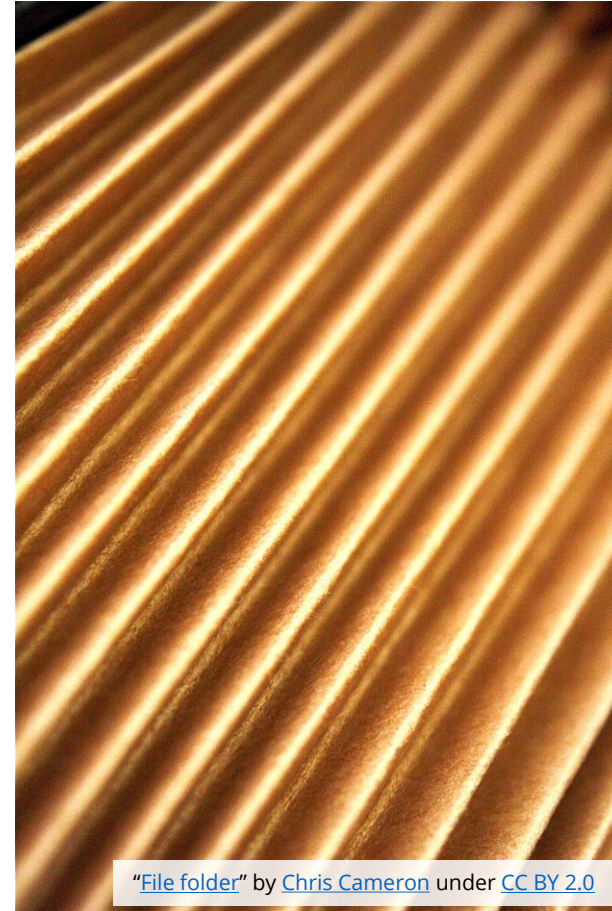
Sketch out possible structures **before implementing**.



"Bibliography" by Alexandre Duret-Lutz under CC BY-SA 2.0

Files and Folders: Example Scheme

- Documents
 - Personal
 - Activities
 - Financial
 - School
 - College
 - Course #1
 - ...
 - Continuing Education
 - Work
 - Job #1
 -



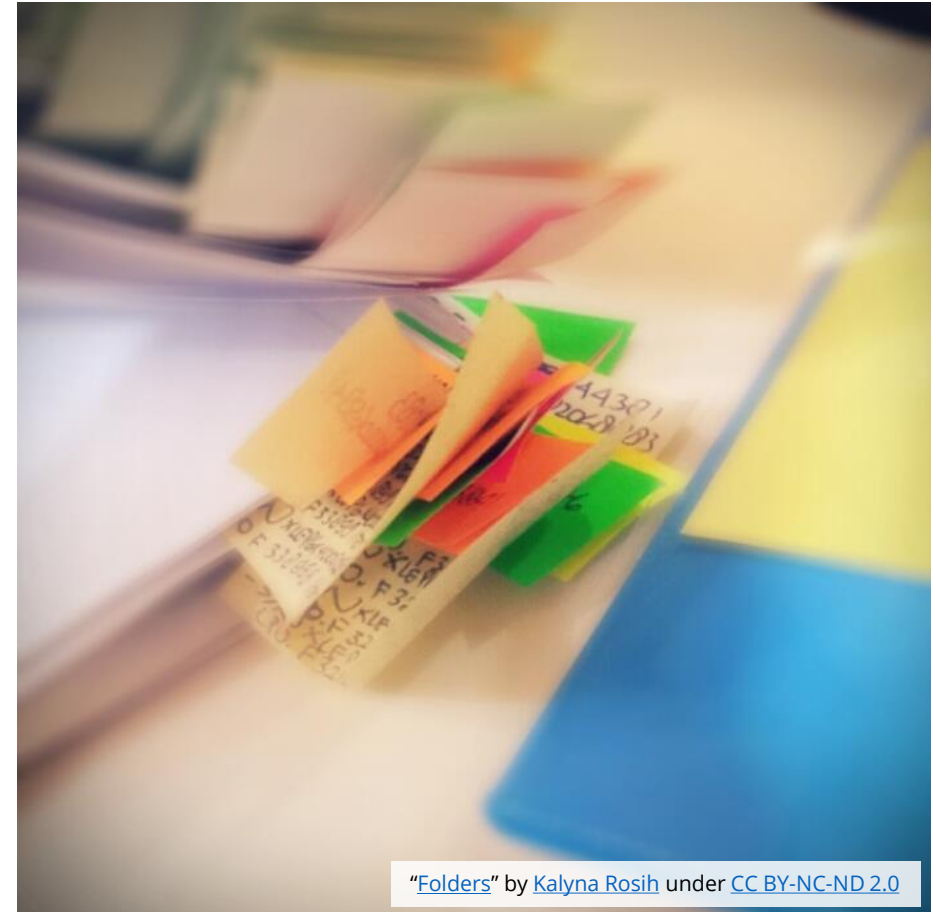
"File folder" by [Chris Cameron](#) under [CC BY 2.0](#)

File Naming

Don't rename files **if you don't have to**.

The default file naming schemes for photos may be obscure, but you're generally going to **access those based on metadata**.

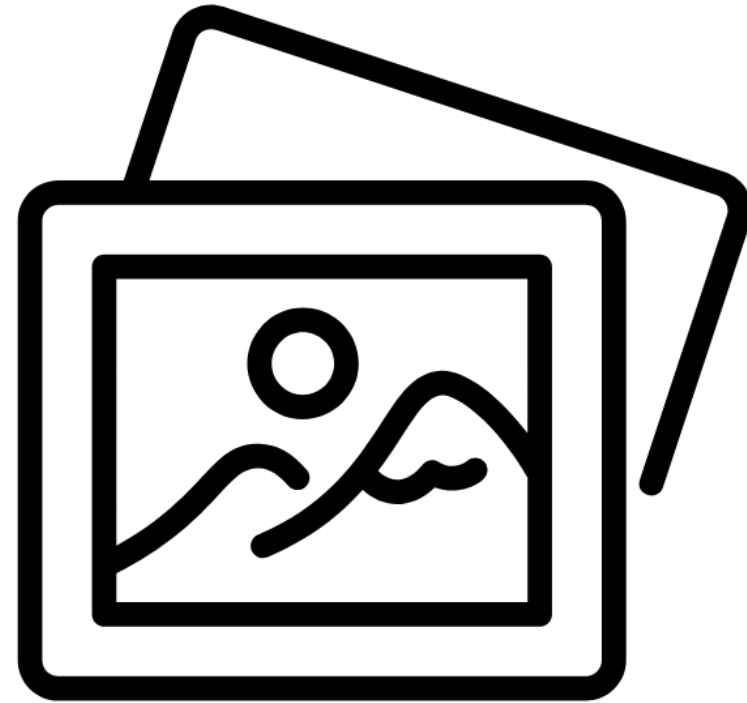
File names matter more if retrieval is based more on **browsing a filesystem** than by searching metadata.



"Folders" by [Kalyna Rosih](#) under [CC BY-NC-ND 2.0](#)

Photo Formats

- JPEG
- TIFF
- HEIC
- RAW



Created by omeneko
from Noun Project

Photo Formats: JPEG

- JPEG
- TIFF
- HEIC
- RAW

Most of **your digital photos are likely JPEGs**, as that's the native format for most mobile phones and digital cameras.

The JPEG format supports "lossy" compression – i.e., file sizes can be reduced by trading off on quality.

Photo Formats: TIFF

- JPEG
- TIFF
- HEIC
- RAW

Scanners sometimes output in TIFF by default. It's also more commonly used as part of print publishing workflows.

TIFF supports "lossy" (ala JPEG) as well as lossless compression (i.e., ala zip files).

Photo Formats: HEIC

- JPEG
- TIFF
- HEIC
- RAW

[HEIC](#) is much like JPEG though with **improved trade-offs between file size and quality** (i.e., relatively better quality at the same level of compression as a comparable JPEG).

HEIC is most commonly used by Apple devices. It is **not as widely supported as JPEG**.

Photo Formats: RAW

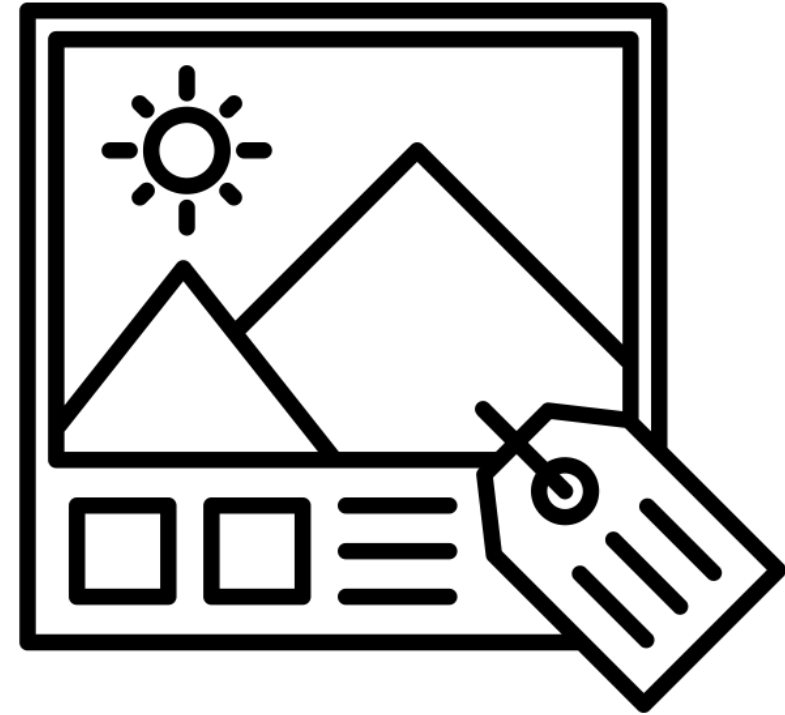
- JPEG
- TIFF
- HEIC
- RAW

[RAW](#) files are **unprocessed digital photos**. They provide greater flexibility for post-processing and re-editing but may be larger in size.

RAW files vary by camera manufacturer and thus **are somewhat non-standard**, limiting generic software support.

Photo Metadata Standards

- EXIF
- IPTC
- XMP



Created by myiconfinder
from Noun Project

Photo Metadata Standards: EXIF

- EXIF
- IPTC
- XMP

[EXIF](#) is a specification for **metadata embedded in digital photos** by digital cameras, including smartphones and scanners.

It includes **timestamps**, camera settings, location info, and more.

EXIF-fielded data is **editable**.

Photo Metadata Standards: IPTC

- EXIF
- IPTC
- XMP

[IPTC](#) is a specification for **descriptive metadata embedded in digital photos.**

There are **few fields in the standard** which limits precision but improves interoperability.

These days, most photo management applications use the **XMP standard instead.**

Photo Metadata Standards: XMP

- EXIF
- IPTC
- XMP

[XMP](#) is another specification for **descriptive metadata** for digital photos, video, and documents.

The standard provides for many fields and is **widely supported**.

Photo Software: ExifTool + ExifToolGUI

[ExifTool](#) is a free command-line utility for **editing digital photo embedded metadata**.

It is **rigorously standards-compliant**, assuring interoperability.

[ExifToolGUI](#) provides a **complementary user interface**, to make ExifTool more usable.



Photo Software: GeoSetter

[GeoSetter](#) is a free, open-source **digital photo geotagging** application.

Even GPS-enabled cameras (e.g., smartphones) **may not be all that precise** in shot placement; GeoSetter lets you correct that after-the-fact.



Photo Software: digiKam

[digiKam](https://digiKam.org) is a free, open-source **comprehensive digital photo management** application.

It supports applicable metadata standards, local AI-powered face tagging, and **hierarchical keyword tagging**, among other features.



Hierarchical Keyword Tagging

Example Keyword Hierarchy

- Location
 - United States of America
 - New Mexico
 - Santa Fe

As Represented in XMP

In the XMP-dc:Subject field, delimited values (i.e., keyword tags) for:

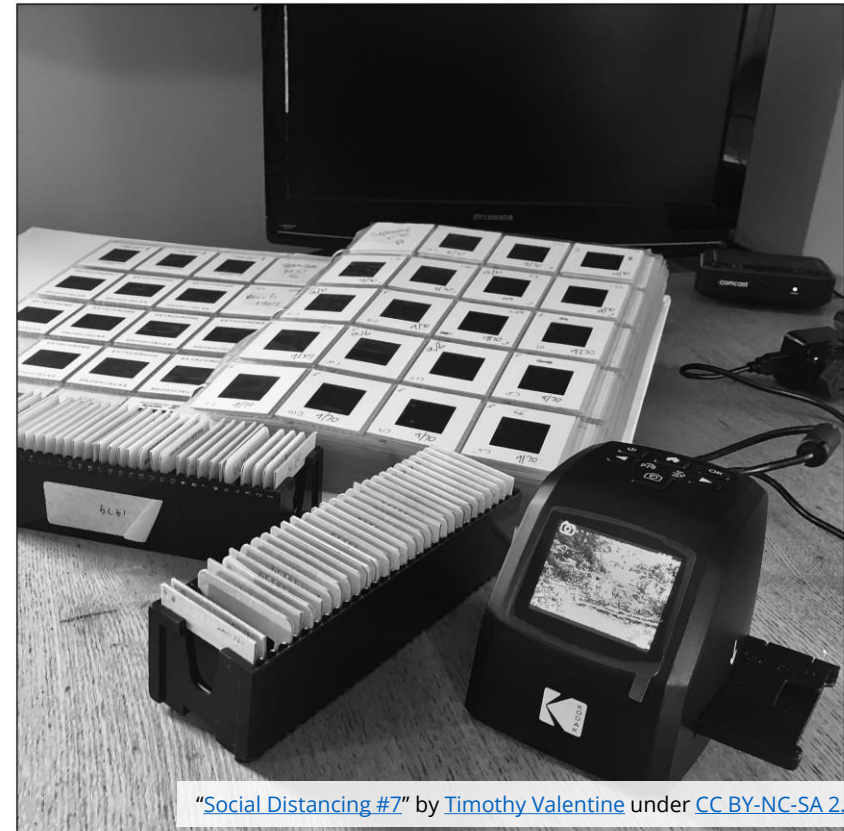
- Location
- United States of America
- New Mexico
- Santa Fe

Photo Software: Other Options

- [Adobe Photoshop Elements](#) – commercial digital photo management software aimed at consumers supporting hierarchical keyword tagging, photo tagging, and editing
- [Adobe Bridge](#) – commercial digital photo management software coupled with Photoshop aimed at professionals
- [Google Photos](#) – free, cloud-based digital photo management platform with flat keyword tagging, face tagging, editing, and social features

Digitization Considerations

- Outsource or DIY?
 - If outsource, send-away service or local provider?
- Comprehensive or selective?
- Relative costs (and kinds of costs) for digitization vs. curation
- Keep the originals?



Considerations for Digitized Photos

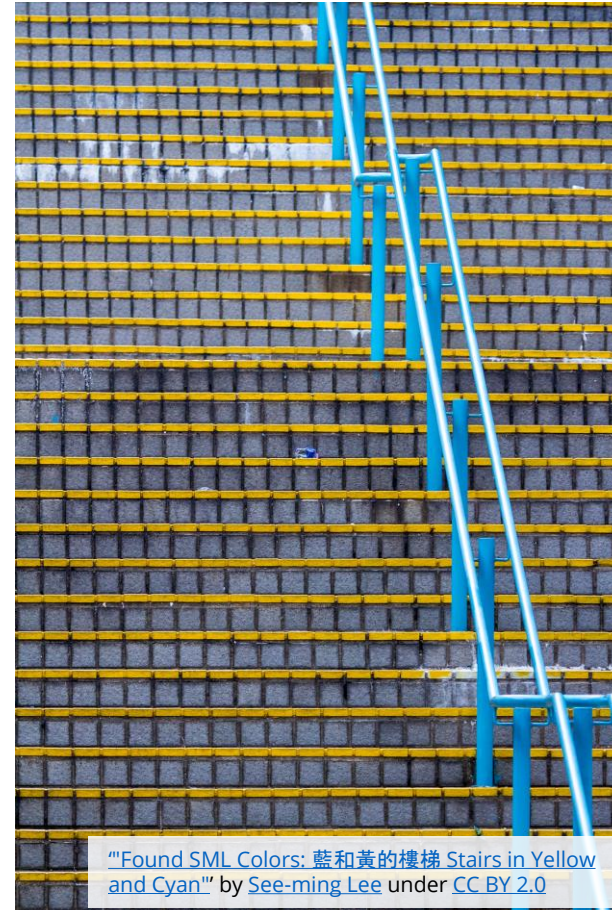
Embedded EXIF timestamps will be those of **when the photo was scanned**, not when it was originally taken.

The color models and tint of old photos are **markers of their era**, but these can also be remediated if desired for aesthetic purposes.



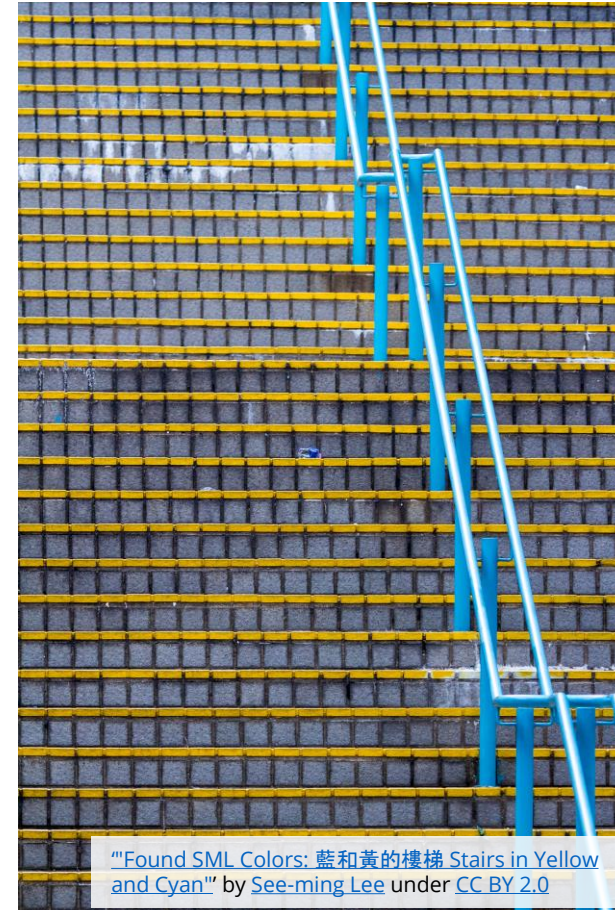
Example Photo Processing Workflow

- ExifTool / ExifToolGUI
 - (If digitized) update
EXIF:DateTimeOriginal
- Geosetter
 - Geotag
- digiKam
 - Face tagging
 - Keyword tagging
 - Assign to album(s)



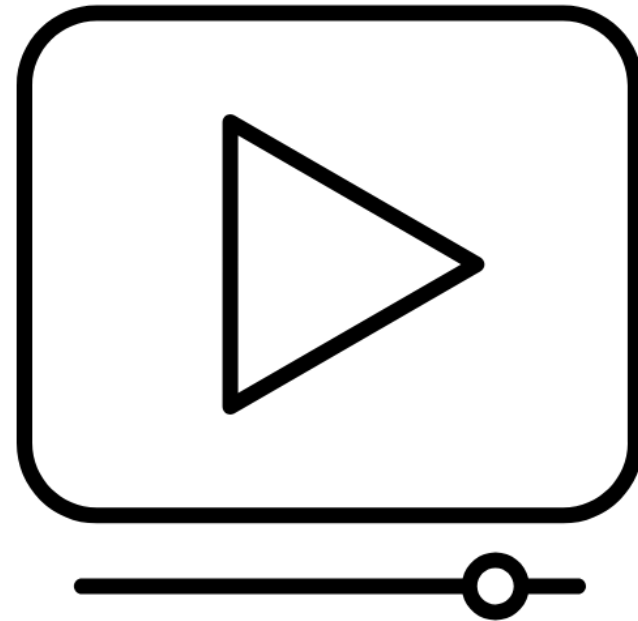
Example Photo Processing Workflow

- Establish canonical data store
- ExifTool / ExifToolGUI
 - (If digitized) update EXIF:DateTimeOriginal
- Geosetter
 - Geotag
- digiKam
 - Face tagging
 - Keyword tagging
 - Assign to album(s)
- Make controlled copies



Video Formats

- MOV
- MP4



Created by Rahmad romadoni
from Noun Project

Video Formats: MOV

- MOV
- MP4

[MOV](#), also known as the QuickTime File Format, is a **multimedia (i.e., video plus audio) format**, most commonly used by Apple devices.

Video Formats: MP4

- MOV
- MP4

[MP4](#) is a **widely used and supported** multimedia (i.e., video plus audio) format.

Most contemporary cameras and digitization processes **will yield MP4** for video output.

E-Mail Services

Many folks access their e-mail via **webmail or mobile app**. In either of these cases, the e-mails live on the server.

Some services allow **access through e-mail clients** using [IMAP](#) or [POP](#) protocols.

IMAP syncs the e-mail store locally; POP downloads e-mail and removes it from the server.



"Mailboxes" by Alexander Baxevanis under CC BY 2.0

E-Mail Formats

- EML
- MBOX
- MSG
- PST



Created by Damar Creative
from Noun Project

E-Mail Formats: EML

- EML
- MBOX
- MSG
- PST

[EML](#) is a **generic format for individual messages** saved outside of an e-mail client.

EML files are generally legible in any software that can read plain-text content.

E-Mail Formats: MBOX

- EML
- MBOX
- MSG
- PST

MBOX is a **generic format for storing e-mail messages.**

For some e-mail clients, it's the native back-end storage format. Many more clients export to MBOX.

Messages are stored in MBOX files **concatenated and in plain-text.**

E-Mail Formats: MSG

- EML
- MBOX
- MSG
- PST

[MSG](#) is a **Microsoft Outlook-specific format** for individual messages saved outside of the mailbox.

Files **may need to be converted** before they can be read or imported by third-party e-mail software.

E-Mail Formats: PST

- EML
- MBOX
- MSG
- PST

[PST](#) is a **Microsoft Outlook-specific format** for storing e-mail messages and calendar events.

Files **may need to be converted** before they can be read or imported by third-party e-mail software.

E-Mail Metadata

Most affordances for curating e-mail are **software-specific**.

Typical options include **organizing messages into folders or tagging**.

These changes **may or may not be portable** to other software.

E-mail otherwise **natively has lots of metadata** – e.g., sender, recipient(s), timestamp, subject; it's just a matter of harnessing it.



Created by Royyan Wijaya
from Noun Project

E-Mail Software: Thunderbird

[Thunderbird](#) is a free, **open-source e-mail client**.

It supports the primary e-mail protocols. It can also gain **additional features and capabilities** through add-ons.

Messages are **stored in MBOX**.



E-Mail Software: ePADD

[ePADD](#) is free, open-source software for **novel exploration of e-mail archives**.

It allows for **analysis and visualization** of correspondent conversations over time, sentiment analysis, entity extraction, and more.



E-Mail Software: Aid4Mail Converter

[Aid4Mail Converter](#) is a commercial **e-mail conversion** and migration application.

It supports **lossless transformation** of data from and to many e-mail formats.



Document Formats

- DOC, DOCX
- XLS, XLSX
- PPT, PPTX
- PDF
- WPD
- WRI
- RTF

If you've authored documents for any appreciable length of time, you likely have **many different types**.

Fortunately, most modern productivity software **can readily handle** most legacy document file formats.

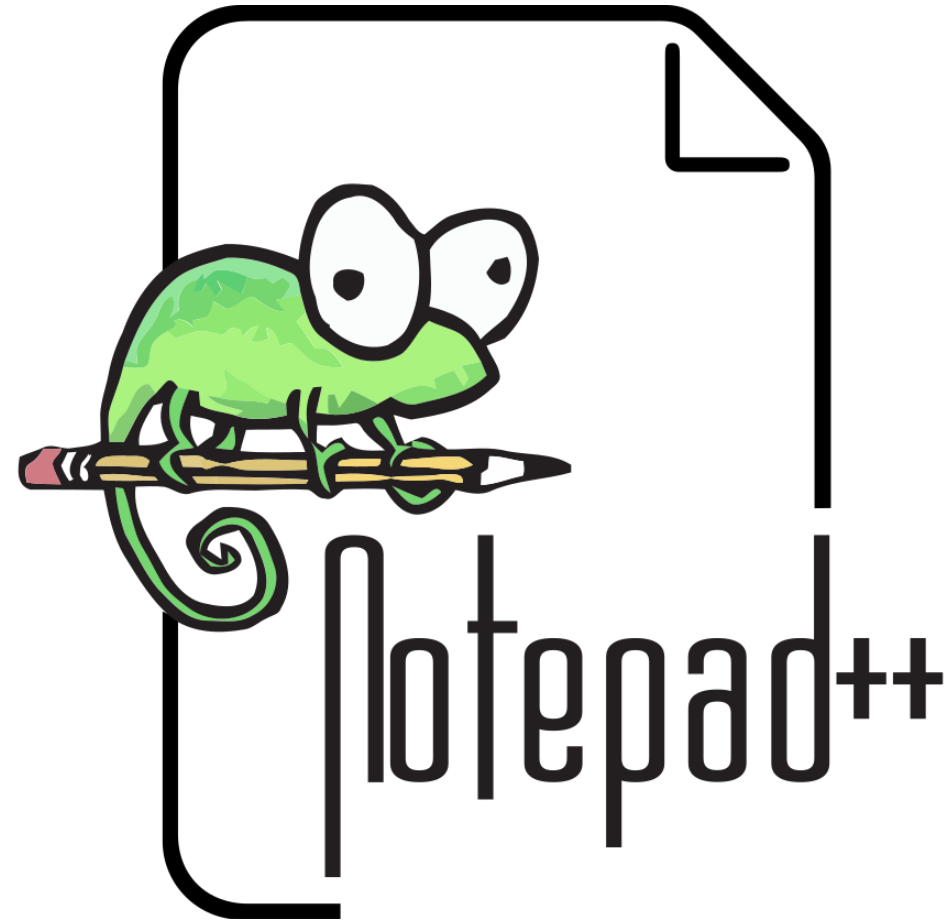
Document Software: LibreOffice

[LibreOffice](#) is a free, open-source office productivity suite with **excellent support for legacy file formats** as well as most other features you'd expect.



Document Software: Notepad++

[Notepad++](#) is a free, open-source text editor that provides for convenient editing of any **structured or unstructured plain-text** file (e.g., XML, software code, etc.).



Genealogy Formats

- GEDCOM

[GEDCOM](#) is the de facto standard for **storing genealogical data** and is widely supported.

The data is technically **structured plain-text** and can thus be read in a text editor.

Genealogy Software: Gramps

[Gramps](#) is free, open-source software for **genealogical documentation**, research, reporting, and analysis.



Ancestry.com Access via SFPL

A quick reminder that we have [access to Ancestry](#) via **the Santa Fe Public Library**.

It's limited to in-library access only, though.

You can **download records for import** into your family tree software.



Accessing Files on Legacy Hardware

Online retailers sell external floppy and optical media drives as well as **adapters** to hook up hard or flash drives.

For more esoteric storage media (e.g., zip disks), you may need to search a marketplace **like eBay**.



Preserve



"The Sounds Of Earth, Voyager Golden Record" by [Kevin Baird](#) under [CC BY-NC-ND 2.0](#)

What do we mean by *preserve*?

“To take action to prevent deterioration or loss.”

– [Dictionary of Archives Terminology, Society of American Archivists](#)



“Engraved in Stone” by [Robert Hruzek](#) under [CC BY-NC-ND 2.0](#)

What do we mean by *preserve*?

In other words, to preserve is to pre-emptively and continually **address risks of data loss**.

Another way of characterizing preservation – that usefully elides the “dark archive” implication – is “**persistent access**”.



"Hug" by [Robert Hruzek](#) under [CC BY-NC-ND 2.0](#)

Long-Term Risks to Data

- Media / hardware failure
- Media / hardware obsolescence
- File format obsolescence
- Operator error
- Natural disaster
- Cybersecurity threat



Media / Hardware Failure

Risk

No removable or fixed storage media lasts forever. It may wear out or be damaged, **rendering its stored data unreadable.**

Mitigation

Make many (inventoried) copies of data, on **different types of storage.**

Media / Hardware Obsolescence

Risk

A suitable reader may no longer be available for legacy removable media or a suitable interface **no longer available** for legacy fixed storage devices.

Mitigation

Forward-migrate data to modern storage media, as early as possible.

Purchase **legacy readers or interface adapters** if needed to recover data.

File Format Obsolescence

Risk

Data stored in esoteric, legacy, and / or proprietary file types is **no longer legible with contemporary software**.

Mitigation

Save data in or export data to **non-proprietary file formats** whenever possible.

Leverage open-source software for robust legacy file format handling.

Operator Error

Risk

You **accidentally and irrevocably delete** or modify the only copy of data that you had intended to keep.

Mitigation

Make **multiple copies of data on decoupled storage**, such that a mistake can't possibly affect all copies at once.

Natural Disaster

Risk

A natural disaster **destroys the media on which data is stored** (i.e., in the process of destroying much else).

Here in Northern New Mexico, we're principally concerned with the risk of **wildfire**.

Mitigation

Make multiple copies of data on **geographically dispersed storage**.

Cybersecurity Threat

Risk

The system on which data is stored is compromised, allowing an attacker to destroy, encrypt, or otherwise **deny access to the data.**

Mitigation

Maintain antivirus and malware protections, and follow best practices.

Make **multiple copies of data on decoupled storage**, such that one compromised system doesn't undermine the integrity of all copies.

Practical Takeaways on Risks

There's **no surefire way** to assure the persistence of any given copy of data.

Don't be cavalier with any given copy, but recognize that fewer copies on coupled systems is **inherently riskier**.

Diversification and redundancy are the best strategies to assure the persistence of at least one copy.



"terracotta warrior" by [uncle.o.](#)
under [CC BY-NC-SA 2.0](#)

Lots of Copies Keep Stuff Safe (LOCKSS)

Avoid single points of failure:

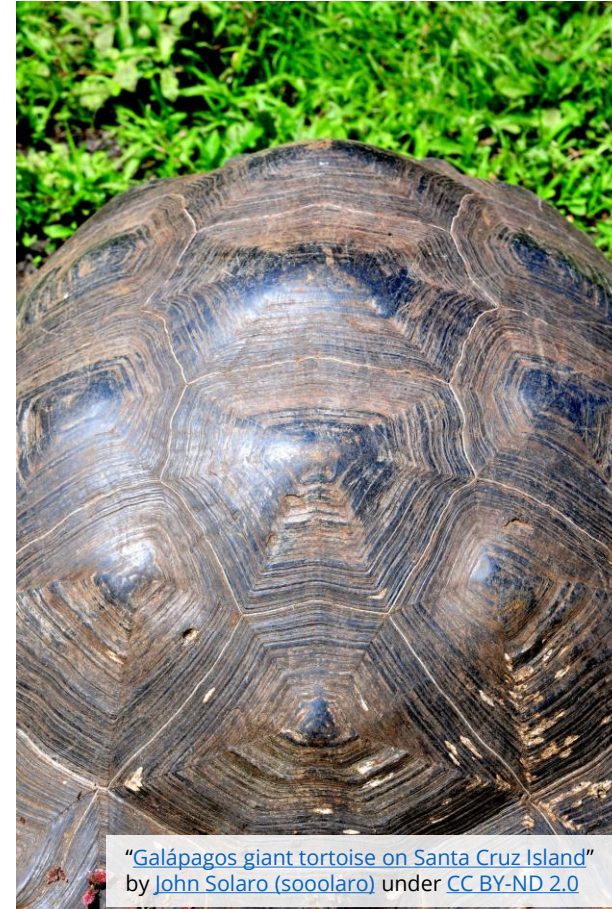
- Only copy or copies hosted by one company
- Only copy or copies stored in one geographic location
- Only copy or copies stored on one computer
- Only copy or copies stored on one hardware device



"Warhol Marilyn" by Joseph Timmons under [CC BY-NC 2.0](#)

Example of a LOCKSS Setup

- Canonical data on local drive
- Make another local copy (e.g., internal / external hard drive, flash drive, optical media)
- Copy to external drive; store in geographically remote location
- Continuous, encrypted online backup (e.g., via Backblaze)
 - secure.backblaze.com/r/027krz





What small, immediate step could you take to better safeguard your personal digital artifacts?

Access





How do you want to be able to retrieve and interact with your personal digital artifacts?

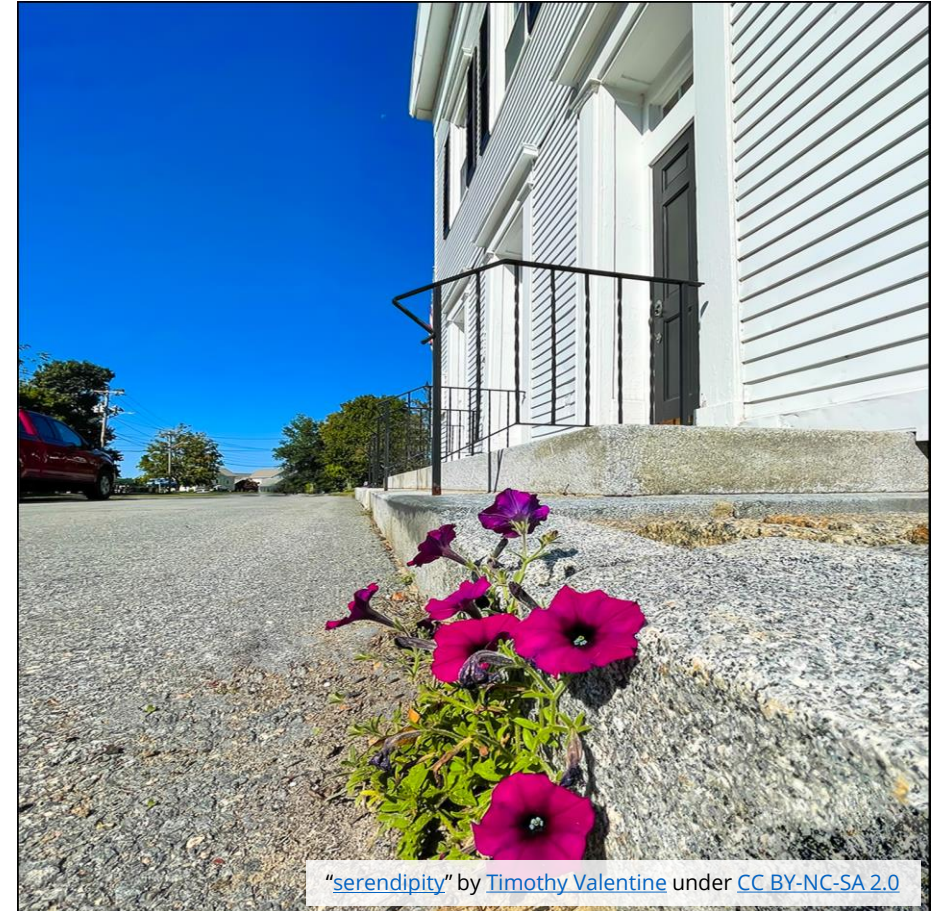
What are my access goals?

I find this the hardest part.

Eschewing cloud services for sharing means **I share less** and miss out on social feedback.

Most of my archive access is thru progressive curation, which also serves as **a memory exercise**.

Precision retrieval is handy, but engineering **serendipitous discovery** is challenging.



"serendipity" by Timothy Valentine under CC BY-NC-SA 2.0

Access Modalities

Search

- Based on metadata
- Based on textual content
- Based on visual content (i.e., via AI indexing)
- Depending on software, may allow for use of logical operators, e.g., AND or OR

Browse

- Based on arrangement of files in folder structures
- Based on indexed file characteristics (e.g., name, timestamp, file type, etc.)

Retrieving Photos w/ Metadata

A decent photo management application (e.g., digiKam) should allow for **arbitrary, logical searches** on a well-curated photo collection, e.g.,

- *Photos w/ all these individuals, within this date range*
- *Photos w/ these keyword tags but not this one*
- *All photos taken within this bounding box drawn on a map*



"The 7 Train @ Shea Stadium" by [bwrabbitgirl](#) under [CC BY-NC-ND 2.0](#)

Content Disposition Instructions

Our archives are ours before they're others'.

We may not want family members to have unfettered access to all of our files.

Consider adding directives for the **disposition of specific content** along with other such instructions.



"Sunday Work" by Rosmarie Voegtli under CC BY 2.0



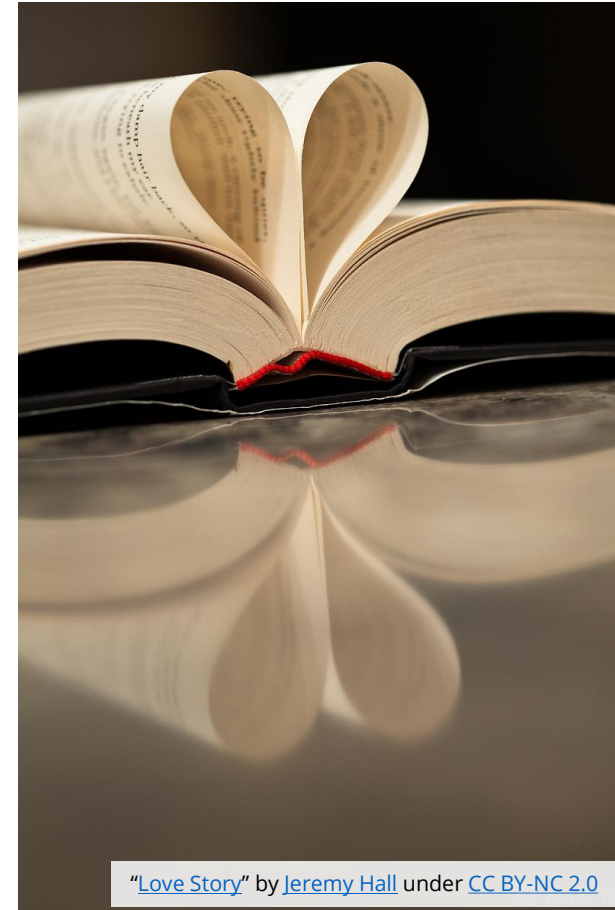
How do you want others to be able to retrieve and interact with your personal digital artifacts?

Storytelling

Don't miss the forest for the trees.

The best atomic metadata
facilitates directed retrieval
around *who, where, and when*.

It is ideally **complemented by storytelling**, helping to answer *why* and *how*, which connect folks with the archive.



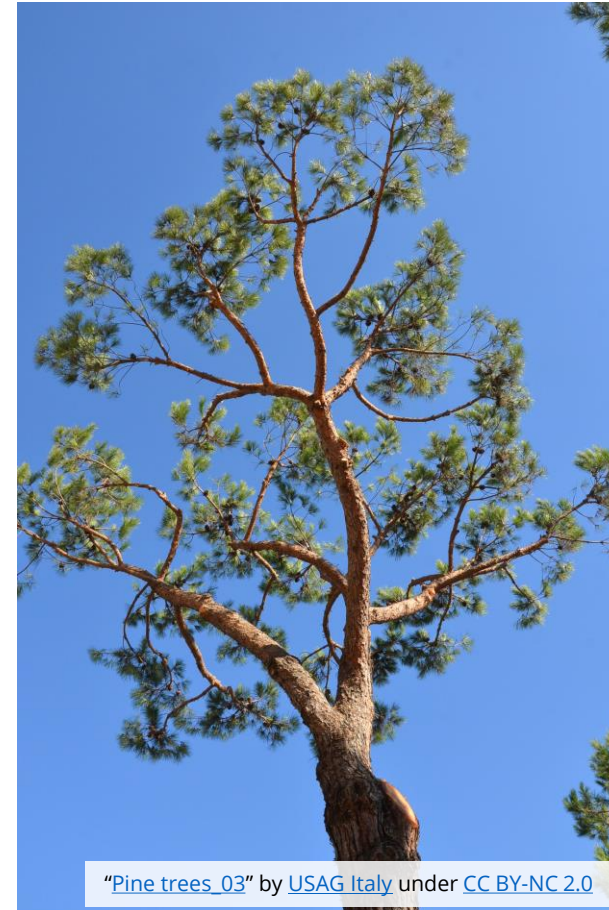
"Love Story" by [Jeremy Hall](#) under [CC BY-NC 2.0](#)

Intergenerational Hand-Offs

It's not in our power to assure the persistence of the archive.

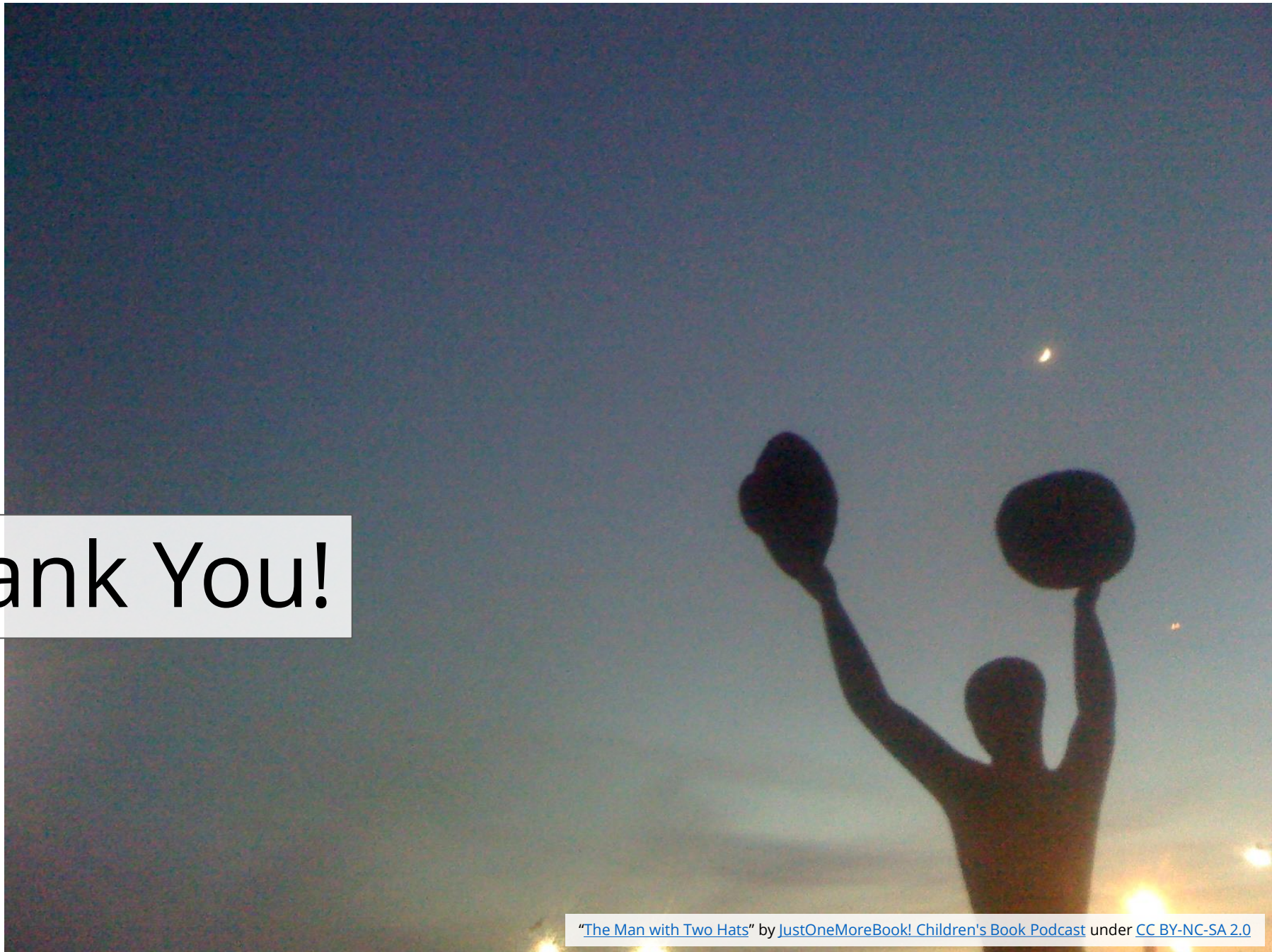
The persistence of the archive depends far more on **its meaning to its inheritors** than on any technical preservation measures.

For that reason, **welcome the next generation's re-appraisal** even if that means they discard parts. As with trees, pruning supports longer-term sustainability and revitalization.



"Pine trees_03" by [USAG Italy](#) under [CC BY-NC 2.0](#)

Thank You!



"The Man with Two Hats" by [JustOneMoreBook!](#) Children's Book Podcast under [CC BY-NC-SA 2.0](#)